End-to-End Max-Min Fairness Optimization for Two-Tier UAV Networks: A Differentiable Joint Trajectory Approach

Yutian Bai, Haiyan Li, Jingyi Ma, and Xuan Li
School of Information and Communication Engineering
Beijing University of Posts and Telecommunications
Beijing, China
baiyutian@bupt.edu.cn, lhy2023@bupt.edu.cn, spencer@bupt.edu.cn, xuan.li@bupt.edu.cn

***Abstract*—The convergence of Sixth Generation (6G) Space-Air-Ground Integrated Networks (SAGIN) and Digital Twin (DT) systems demands ultra-reliable end-to-end data delivery from mobile ground users to the core network. A two-tier Unmanned Aerial Vehicle (UAV) architecture, with access UAVs and backhaul UAVs, provides a scalable solution, yet guaranteeing long-term fairness under user mobility remains an open challenge due to the coupled access-backhaul bottleneck. This paper formulates this problem as maximizing the worst-user cumulative throughput over a time horizon, and introduces Differentiable Joint Trajectory Optimization (DJTO), a fully differentiable framework that replaces hard max-min operations with smooth approximations, enabling gradient-based joint optimization of all UAV positions across all time slots. Comprehensive experiments against Decoupled Greedy (DG) and Model Predictive Control (MPC) baselines show that DJTO improves the poorest user's cumulative throughput by 20.5% over DG and 20.4% over MPC, while spatiotemporal heatmap waterfalls visually confirm the superior end-to-end coverage and fairness.**

***Index Terms*—6G, DT, differentiable optimization, end-to-end fairness, two-tier UAV networks.**

I. Introduction

The Sixth Generation (6G) of wireless communication is expected to realize a fully connected world through Space-Air-Ground Integrated Networks (SAGIN), where aerial platforms such as Unmanned Aerial Vehicles (UAVs) play a pivotal role in extending coverage, enhancing capacity, and providing on-demand connectivity [1]. Concurrently, Digital Twin (DT) technology is emerging as a cornerstone for real-time monitoring and control of physical systems, requiring continuous and reliable data streams from distributed mobile sensors to the cloud [2]. In such contexts, a two-tier UAV relay architecture—where Lower-Tier UAVs (L1) serve as access points for ground users and Upper-Tier UAVs (L2) act as backhaul nodes to the core network—offers a flexible and cost-effective deployment model [3]. However, the dynamic nature of user mobility and the coupled relationship between access and backhaul links create significant challenges in guaranteeing end-to-end fairness over extended operational periods.

This work was supported by the National Natural Science Foundation of China under Grant 62401076. (Corresponding author: Xuan Li.)

Existing research on UAV trajectory optimization has largely focused on single-tier scenarios or instantaneous performance metrics. Many works optimize the positions of access UAVs under the assumption of ideal or static backhaul connectivity, thus ignoring the potential bottleneck caused by limited backhaul capacity [4], [5]. Others employ Model Predictive Control (MPC) or reinforcement learning to adjust trajectories in real time [6], [7], but these approaches typically optimize short-term objectives and may fail to ensure cumulative fairness over a long mission duration. Moreover, the hard max-min utility function commonly used in fairness-oriented designs is non-differentiable, making it difficult to jointly optimize over continuous spatial and temporal dimensions [8].

To bridge these gaps, this paper formulates the problem of long-term max-min fairness in a two-tier UAV network and proposes a solution with the following contributions:

- Problem Formulation. We model the end-to-end throughput of each user as the bottleneck of its access and backhaul links, and define the optimization objective as maximizing the minimum cumulative throughput over a finite time horizon, subject to UAV mobility constraints.
- Differentiable Joint Trajectory Optimization (DJTO). By replacing hard max and min operations with smooth Log-Sum-Exp (LSE) approximations [9], [10], we construct a fully differentiable loss function that allows gradient-based joint optimization of all UAV positions across all time slots. This framework, implemented in PyTorch, efficiently handles the coupled spatial-temporal variables.
- Comprehensive Evaluation. We compare DJTO with Decoupled Greedy (DG) and Model Predictive Control (MPC) baselines that jointly optimize both UAV tiers under the same speed penalty [11]. Our method improves the poorest user's cumulative throughput by 20.5% over DG and 20.4% over MPC on average.
- Visual Analytics. We introduce waterfall-style spatiotemporal heatmaps that present the evolution of bottleneck rate across space and time, offering intuitive visual evidence of the dynamic coverage and fairness achieved by DJTO.